Backup/Recovery

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ISIN 325
February 19, 2026
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Into/Overview

For this assignment we were given the task to create backup/recovery Plan for the given database environment:

“You are the lead DBA (of 3) of a corporation that has approximately 1500 employees with net sales of $250 million that are expected to grow at 10% per year. Your corporate standard is MS SQL Server, but your HR system is cloud based through a third party cloud provider. Create a five part (section) backup and recovery plan for all the database in your shop including the cloud based HR system. Describe how the in-house vs. cloud based backup/recovery plans are similar and how they differ. The SQL Server databases include legacy systems such as accounting, forecasting, planning, and transactional systems such as order entry and fulfillment, as well as analytical data warehouses”.

This assignment required me to develop a security strategy which would safeguard the company's future growth needs rather than merely creating backups. The company needs to secure its SQL Server systems and cloud based HR system because it generates $250 million in sales and expects yearly growth which requires protection against data loss and downtime and disasters. Additionally, I had to take the perspective of business continuity and recovery time objectives and recovery point objectives while I learned about the cloud shared responsibility model. The comparison between in house systems and cloud environments enabled me to build a complete backup and recovery system which ensures operational stability while supporting future growth.

Backup Schedule and Operations

Schedule 1 (Robust/Expensive)

Day	**SQL Server Systems**
Monday	Full backup at 7 AM, log backups every 30 minutes
Tuesday	Full backup at 7 AM, log backups every 30 minutes
Wednesday	Full backup at 7 AM, log backups every 30 minutes
Thursday	Full backup at 7 AM, log backups every 30 minutes
Friday	Full backup at 7 AM, log backups every 30 minutes
Saturday	Full backup at 7 AM, log backups every 30 minutes
Sunday	Full backup at 7 AM, log backups every 30 minutes

Overview:
This plan is still the most expensive because backups run daily and log backups happen frequently. The timing is consistent each night at 10 PM, which simplifies management while still keeping recovery risk very low.

Schedule 2 (Intermediate/Mid Range)

Day	**SQL Server Systems**
Monday	Full backup at 7 AM, log backups every 5 hours
Tuesday	-
Wednesday	Full backup at 7 AM, log backups every 5 hours
Thursday	-
Friday	Full backup at 7 AM, log backups every 5 hours
Saturday	-
Sunday	Full backup at 7 AM, log backups every 5 hours

Overview:
This plan reduces cost by spacing log backups every 5 hours instead of every 30 minutes. Additionally, we reduced the frequency of full backups and log backups from 7 times a week to 4. All backups still run at 10 PM for consistency. This lowers storage and processing demands while still offering solid recovery capability.

Schedule 3 (Simple/Cheap)

Day	**SQL Server Systems**
Monday	
Tuesday	
Wednesday	Full backup at 7 AM, one time log back up at 5 PM
Thursday	
Friday	
Saturday	
Sunday	Full backup at 7 AM, one time log back up at 5 PM

Overview:
This is the simplest and cheapest schedule. Backups only run at 7 AM on the days they are scheduled, and log backups happen just once per day. This reduces cost and storage needs but increases the chance of data loss and longer recovery times.

Support Protocols

Contact Information and Method

The DBA team has three members, including the Lead DBA. One DBA will be assigned as the primary on call contact each week, with another as backup. The Help Desk will create a high priority ticket to report database problems which will require them to contact the on-call DBA through phone and text messages. The Lead DBA and IT Manager will receive an escalation of the issue when staff members fail to resolve it within a specific time frame. The vendor will be reached through their 24 hour support line or support portal for assistance with the cloud based HR system. The IT operations manual will contain all contact information for vendors and details about their accounts.

Prioritization of Systems

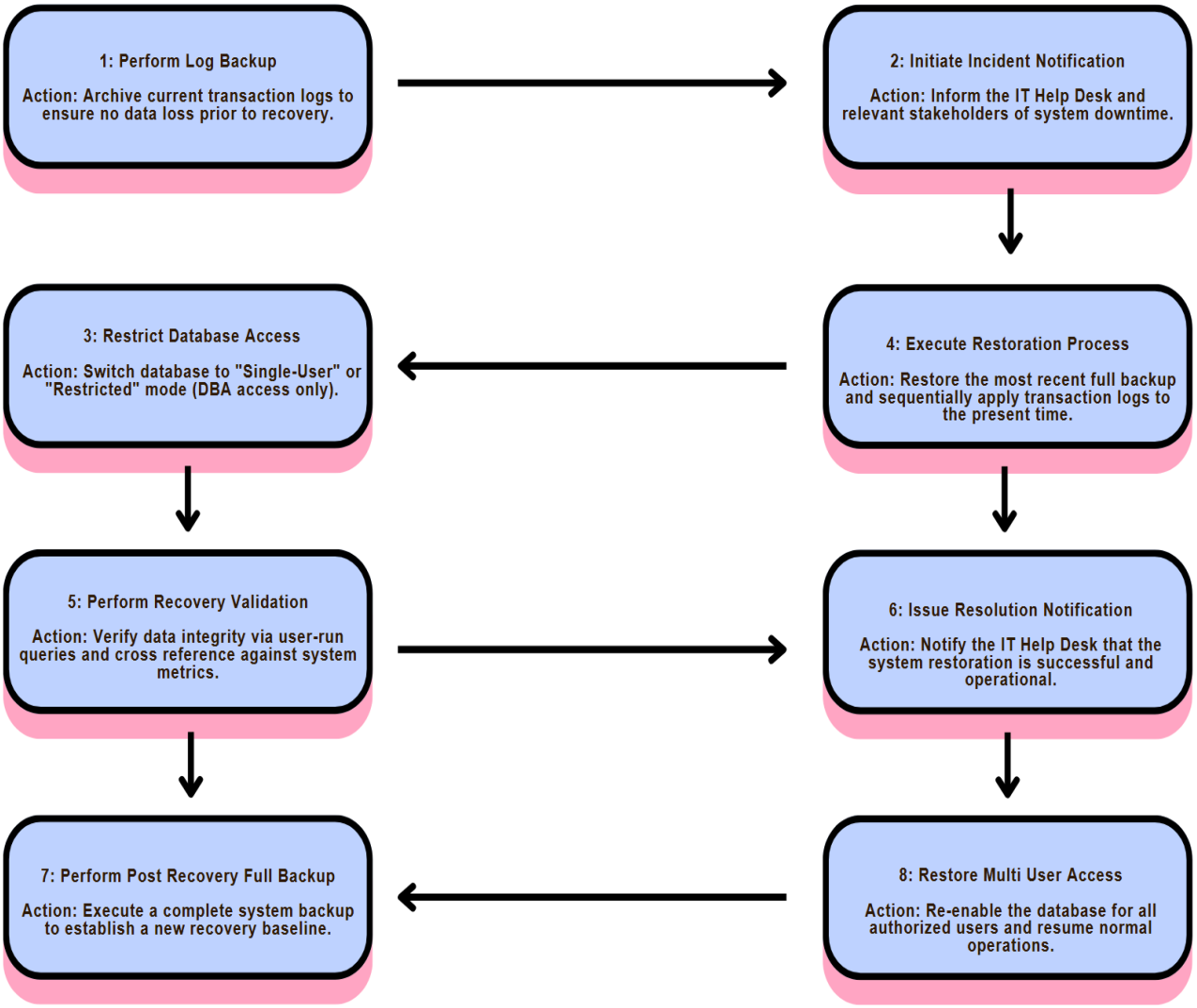
These systems will be restored based on business importance.

- Priority 1: Order entry, accounting, and payroll functions within the cloud HR system. These directly impact revenue and employee pay.
- Priority 2: Forecasting, planning, and reporting systems.
- Priority 3: Analytical data warehouses and historical systems.

Backup for People and Staffing

The organization will establish complete documentation of all backup and recovery procedures which will be kept in a protected shared space. The organization will provide recovery procedure training to at least two DBAs because we want to avoid having a single point of dependency. A trained team member will execute the recovery process in case the database administrator is not accessible. Lastly, the organization and staffers can reach out to outside resources or their cloud HR vendor for assistance when/if they require extra support.

Recovery Process



Metrics

1. Backup Success Rate
Measures how many scheduled backups complete without errors.
Goal: 98 to 100 percent success.

2. Backup Completion Time
Measures whether backups finish within the planned time window.
Goal: All backups complete before business hours.

3. Restore Success Rate
Measures how many test restores are completed successfully.
Goal: 100 percent successful test restores.

4. Recovery Time
Measures how long it takes to restore a system after failure.
Goal: Meet recovery time targets for Priority 1 systems.

5. Data Loss Amount
Measures how much data is lost during recovery.
Goal: Data loss stays within the planned backup interval.

6. Storage Usage
Tracks how much storage backup files are using.
Goal: Keep usage below 80 percent of capacity.

7. Backup Verification
Measures whether backups are tested or verified for integrity.
Goal: 100 percent of backups verified.

8. Incident Response Time
Measures how quickly the DBA team responds to a failure.
Goal: Response within 15 minutes for critical systems.

9. Disaster Recovery Testing Frequency
Tracks how often full recovery tests are performed.
Goal: At least once or twice per year.

10. Documentation Accuracy
Measures whether backup and recovery procedures are current and easy to follow.
Goal: All documentation reviewed and updated at least once per year.

Turnover Process

Simple Change Procedure

A basic change requires index creation and stored procedure update and report modification and minor configuration change. The Lead DBA needs to review the Change Request Form which requires submission for simple changes. The change will receive scheduling after the maintenance window which occurs every week during its approved time. A verified backup must be completed before applying the change which allows for rollback if necessary. The system will undergo testing after implementation to verify correct operation and all results will be documented.

Complex Change Procedure

The changes need a complete Change Request Form together with a risk assessment and impact assessment. The Lead DBA and IT Manager both need to give their approval before a person can schedule the event. The organization will execute complex changes during extended maintenance windows which usually operate on weekends after business hours to minimize their effect on operations. The implementation needs validation testing to be finished before creating the Change Completion Report.

Backups and the Sandwich Concept

For complex changes, the sandwich method will be used. The process requires two full backups which include one backup taken before the change starts and the second backup taken after the change has been completed. The first backup protects the organization from potential rollback needs while the second backup records the system's current stable condition. The minimum requirement for simple changes mandates pre change backup which must be completed before any other work begins. The cloud based HR system requires confirmation of provider backup status before any approved changes can proceed.

Required Forms

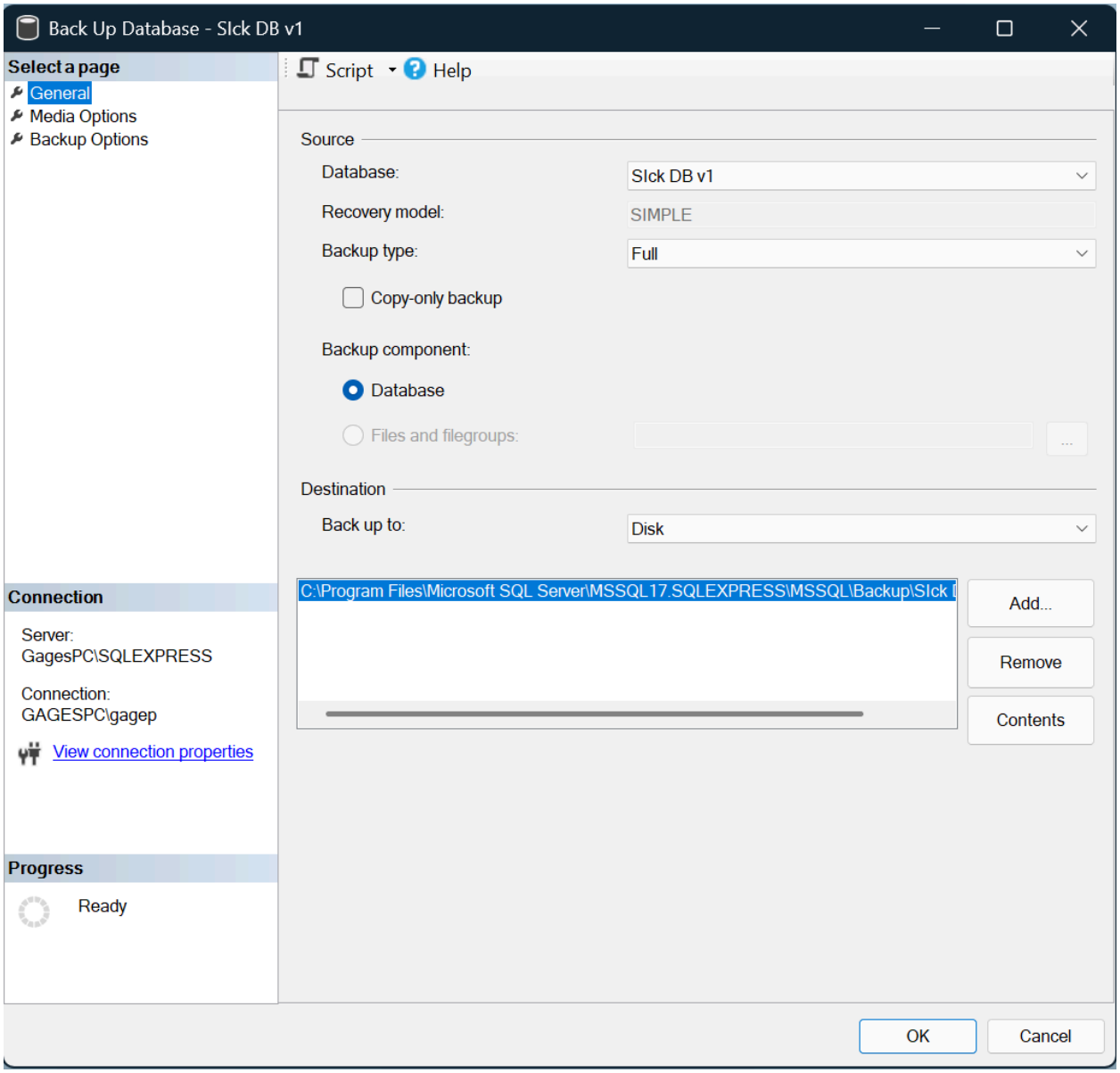
The Change Request Form will include the date of request, system affected, type of change, description of the change, risk level, requested implementation date, and required approval signatures. The Backup Verification Form will document the date and time of the backup, the backup type, verification status, and DBA signature. The Change Completion Report will include the implementation date and time, the outcome

of the change implementation, all problems that arose during the process, proof of post change backup completion, and all final approval signatures.

Timing of Turnover

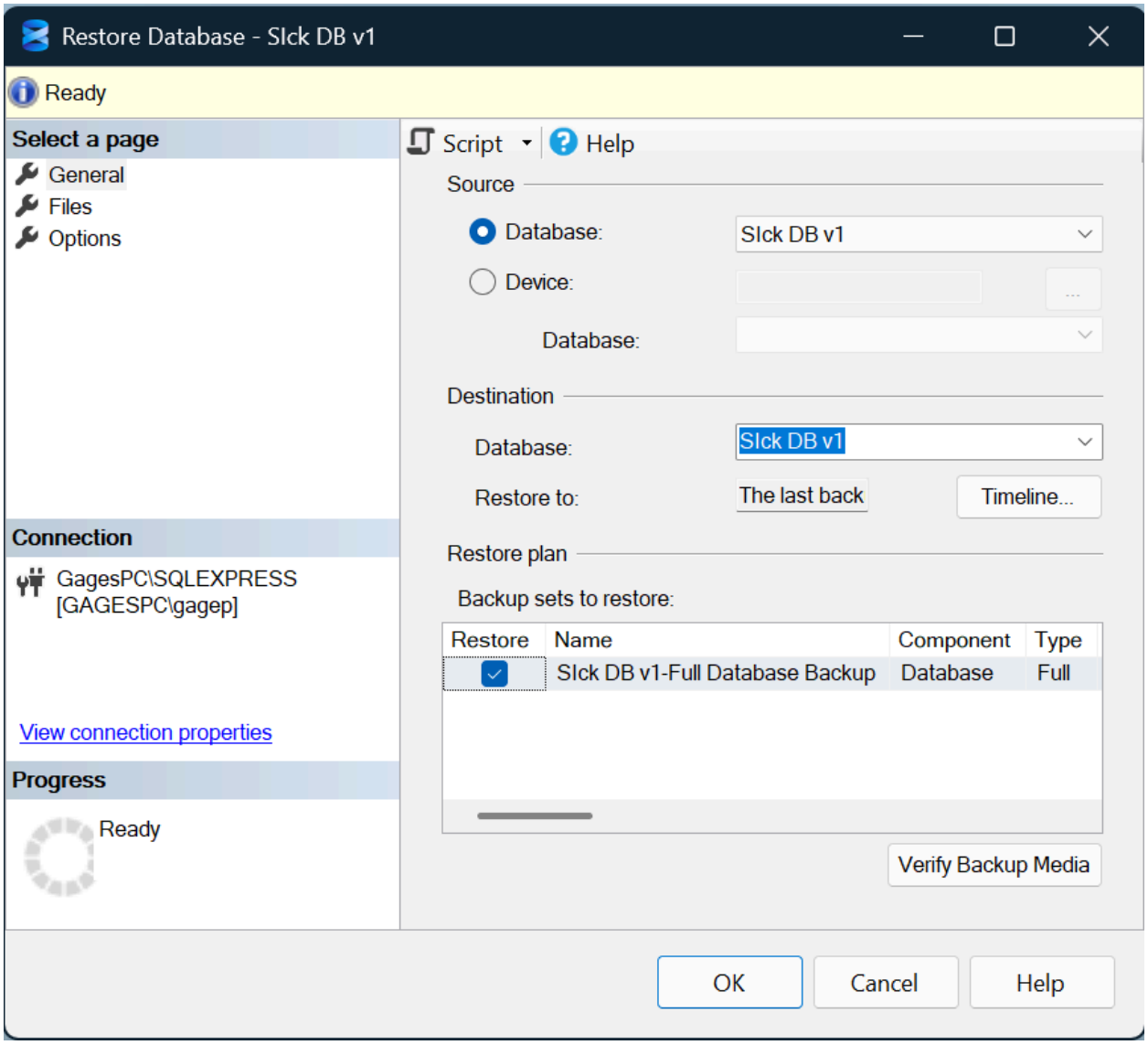
Simple changes will occur during weekly maintenance windows. When complex changes occur, they will be dealt with during approved extended weekend maintenance windows. Emergency changes may occur outside of these windows if necessary, but a pre change backup must still be completed whenever possible.

SQL Backup/Restore (Wizard)



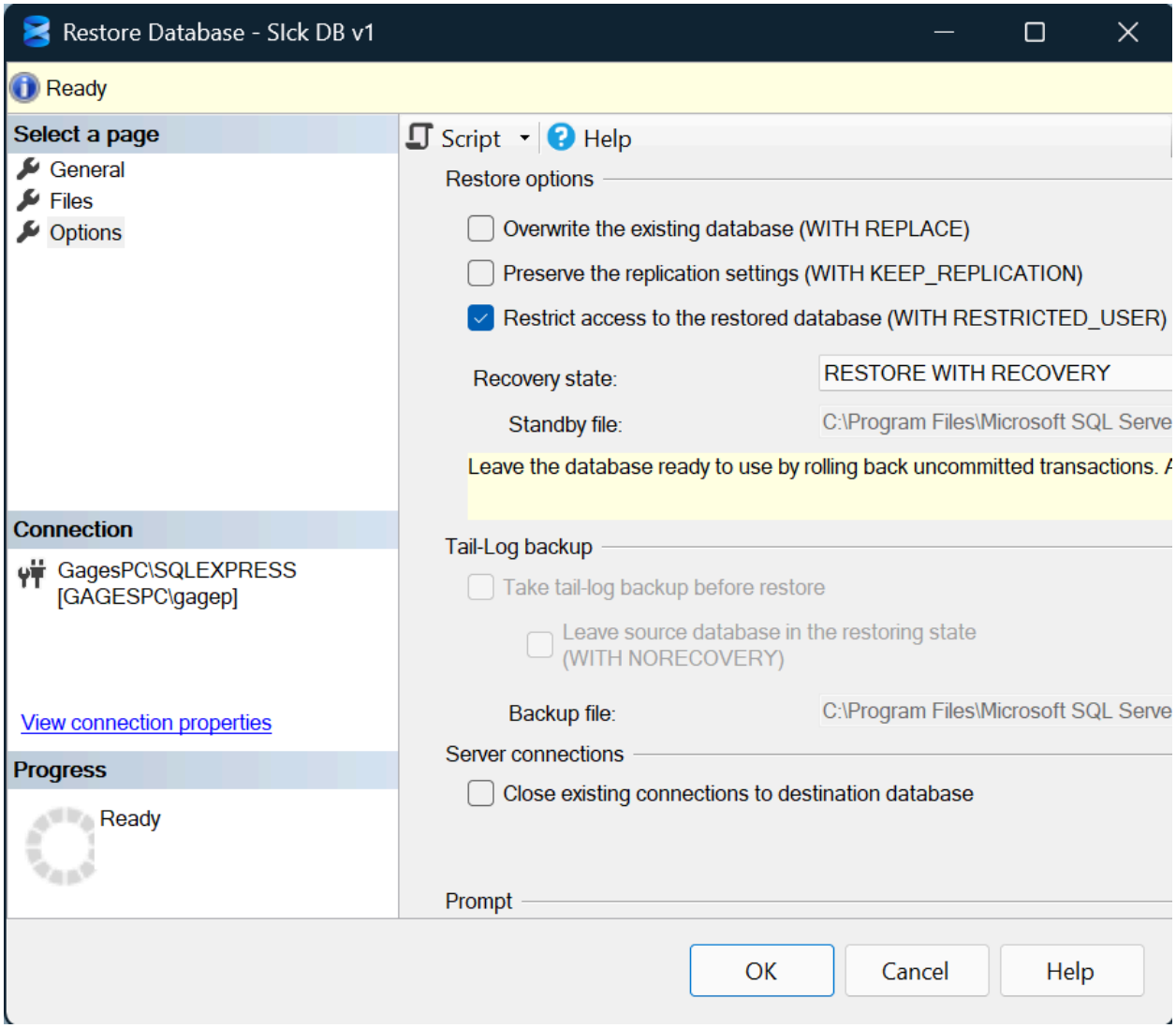
Comments:

To access the wizard we locate our database, right click, tasks, and click on back up. After our previous steps we get this screen. From here we take the defaults and our database has been properly backed up.



Comments:

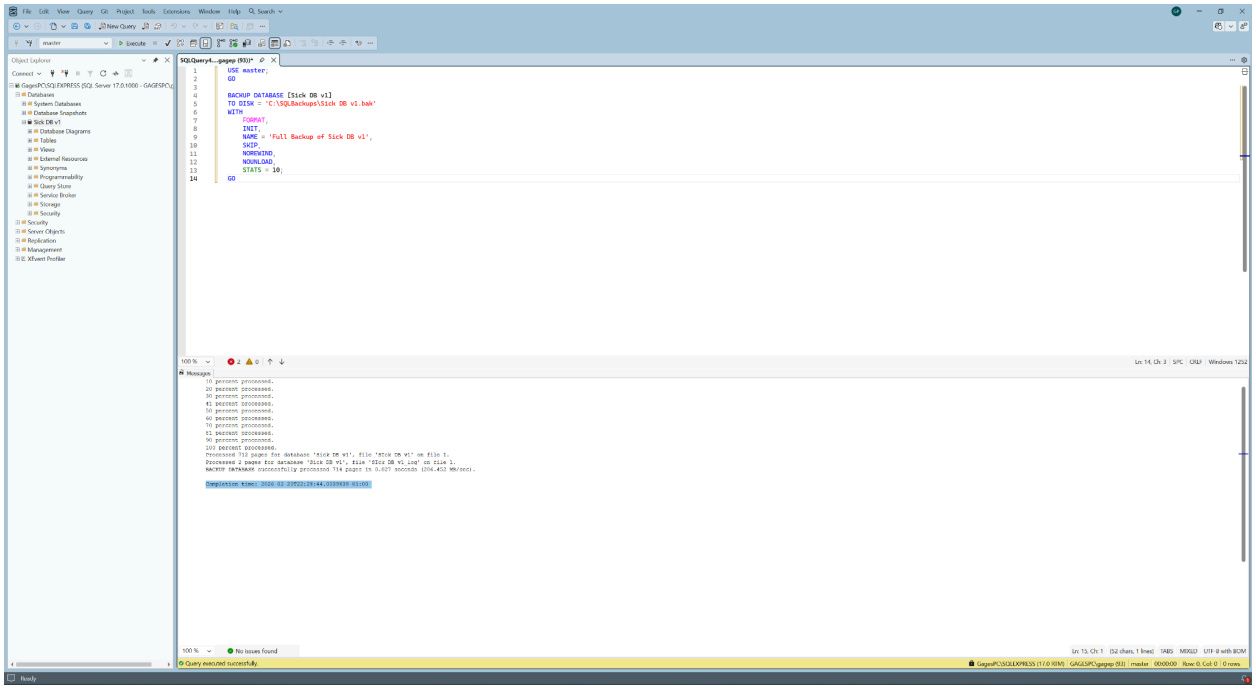
To do a restore we start with the same steps as a back up, locate our database, right click, restore. After our previous steps we get the menu as seen above we take the defaults and we then have successfully restored our database. One important note here is our database will fail to properly restore if we have connections to the database. To find out what connections are in use we can use a sp_who to see what connections are in our database. Once we find what connection it is we can then use a Kill script to break those connections.



Comments:

If we need to put a database into DBO use only mode we use the same steps to reach our restore menu then select options on the lefthand side and check mark “Restrict access to the restored database”.

SQL Backup/Restore (Script)



Comments:

This script created a full backup of the SIck DB v1 in SQL Server Management Studio. It backed up the entire database to a specified disk location in the C:\SQLBackups folder and saved it as a .bak file. The script used options such as FORMAT and INIT to overwrite any existing backup file with the same name and included the STATS option to display progress during execution. This backup ensures that a complete copy of the database is available for recovery in case of system failure, data corruption, or accidental data loss.